

PORTFOLIO OF EVIDENCE

Cloud Infrastructure Deployment

on Google Cloud Platform

C a n d i d a t e

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D a t e

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C o m p e t e n c y L e v e l

Associate — Cloud Infrastructure & Systems Engineering

Project Video Links at the late Page.

Google Cloud Platform

Docker

Linux / Ubuntu

RDP / XRDP

Cloud Engineering · Systems Administration · DevOps

01 PROJECT OVERVIEW

Objective

This project demonstrates the end-to-end provisioning and configuration of a cloud-hosted virtual machine on Google Cloud Platform (GCP), deployed as a fully functional remote working environment. The objective was to design, build, and validate an infrastructure stack capable of hosting both a containerized workflow automation platform and a native AI-assisted development environment on a single cloud compute instance — accessible remotely from any Windows workstation via the Remote Desktop Protocol (RDP).

Scope & Context

The project was executed within a live GCP environment utilizing \$300 in free trial credits, with all configurations applied to real cloud infrastructure. The target environment was an e2-medium compute instance (2 vCPU, 4GB RAM) running Ubuntu 22.04 LTS, hosted in the europe-west2 (London) region — strategically selected to minimize network latency for UK-based business operations.

Technologies Deployed

Technology	Purpose
Google Cloud Platform	Cloud infrastructure provider — VM provisioning, VPC networking, firewall management
Ubuntu 22.04 LTS	Server operating system — stable, long-term support Linux distribution
XFCE + XRDP	Lightweight desktop environment with Remote Desktop Protocol daemon for remote GUI access
Docker	Container runtime for deploying and managing the N8N automation application
N8N Community Edition	Open-source workflow automation platform — self-hosted, no subscription required
Antigravity IDE	AI-assisted development environment installed natively on the remote desktop

Outcome

The project was completed successfully, resulting in a fully operational cloud workstation accessible via RDP, with N8N community edition running as a Docker container on port 5678, and Antigravity IDE authenticated and ready for use. All seven core tasks were completed end-to-end, with real-time troubleshooting applied when firewall and zone availability issues arose. The skills demonstrated span cloud engineering, Linux systems administration, and DevOps practices at an associate competency level.

02 EVIDENCE LOG — SCREENSHOT ANALYSIS

The following figures present documented evidence of each phase of the deployment project. Each screenshot is accompanied by a professional technical caption describing the activity performed, the commands or configurations applied, and the significance of the output within the overall infrastructure build.

Figure 1: Accessing the GCP Billing Portal

Google Cloud Platform (GCP) Billing Overview confirming the activation of the \$300 free trial credit with 84 days remaining. This validates that the project was executed within a live cloud environment using real infrastructure credits, not a simulated or sandbox environment. The billing dashboard confirms zero spend at this point, prior to resource provisioning.

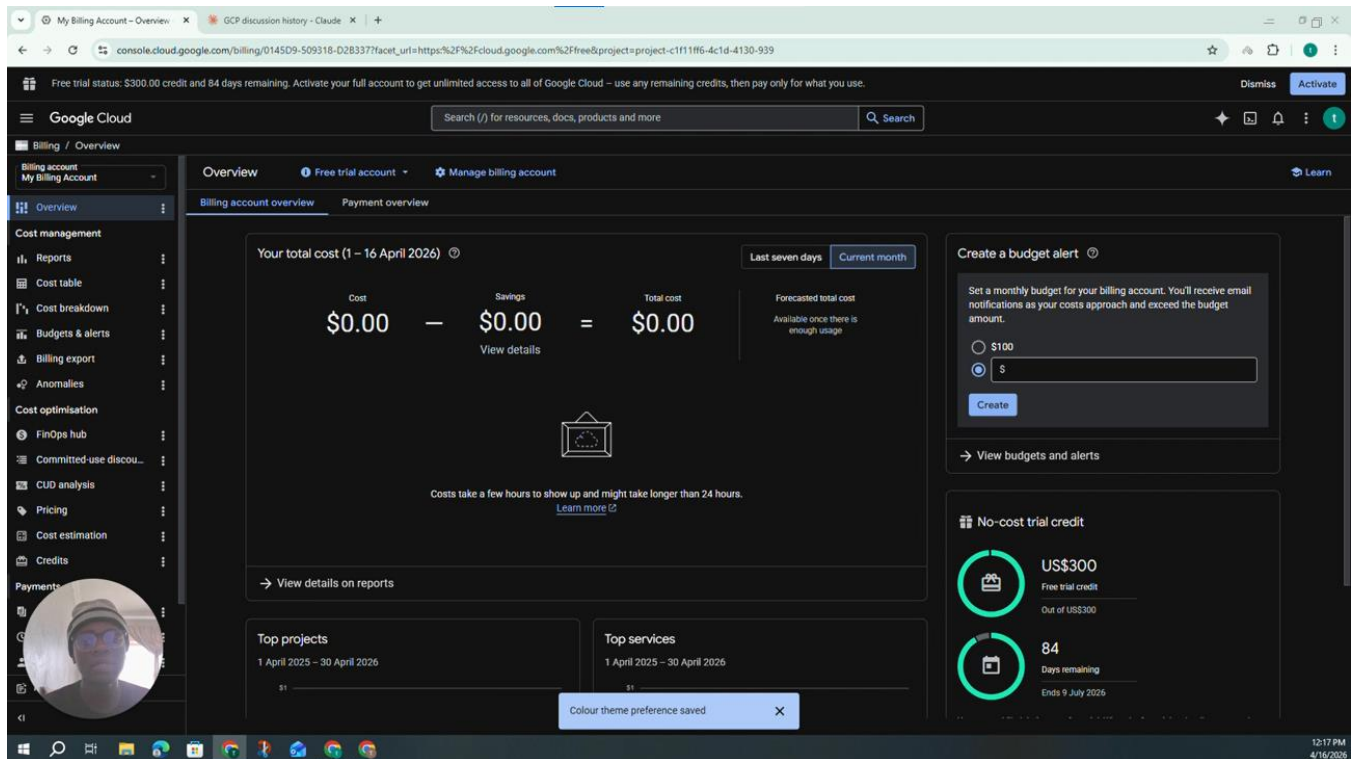


Figure 1

Figure 2: VM Instance Creation — Machine Configuration

Compute instance configuration screen on GCP. Machine type selected: e2-medium (2 vCPU, 4GB RAM), Region: europe-west2 (London), estimated cost: US\$32.71/month. The London region was strategically selected to minimize latency for UK-based client operations. The GCP cost estimator on the right panel confirms the monthly spend projection.

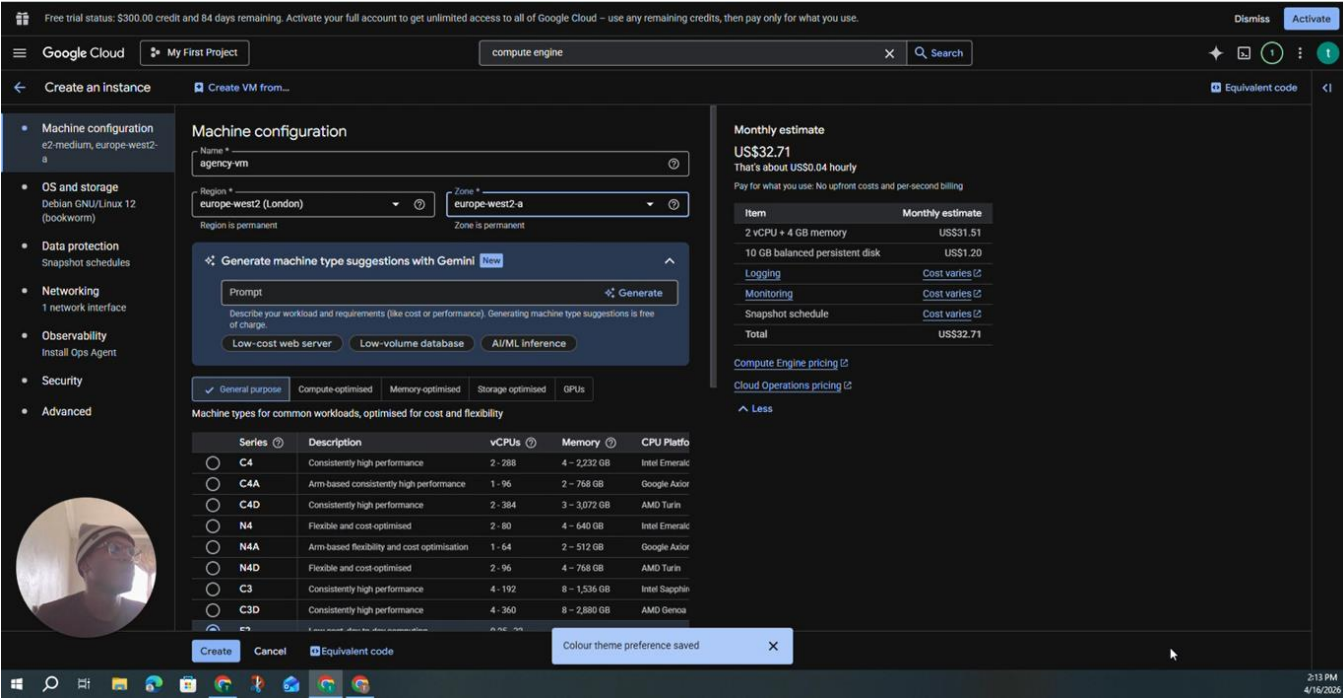


Figure 2

Figure 3: Boot Disk Configuration

Boot disk configuration panel showing the operating system and storage parameters applied to the VM. OS: Ubuntu 22.04 LTS (x86/64, amd64 jammy image), Boot disk type: Balanced Persistent Disk, Storage capacity: 50GB. Ubuntu 22.04 LTS was selected for its long-term support lifecycle, stability, and compatibility with the planned software stack including Docker, XFCE, and XRDP.

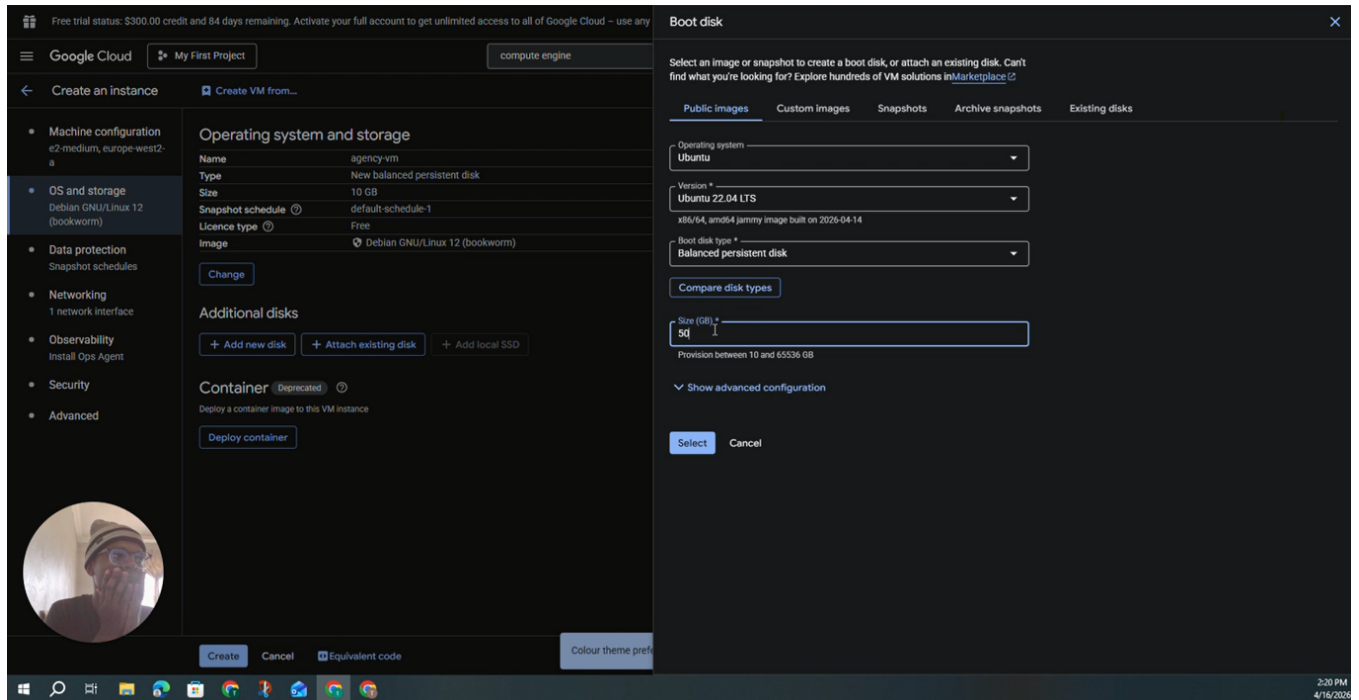


Figure 3

Figure 4: Network and Firewall Configuration

Networking section of the VM creation form, enabling inbound HTTP and HTTPS traffic rules. Network tags http-server and https-server were applied to allow web traffic to the instance — a prerequisite for browser-accessible services such as the N8N automation dashboard. The network interface is configured on the default VPC (10.154.0.0/20).

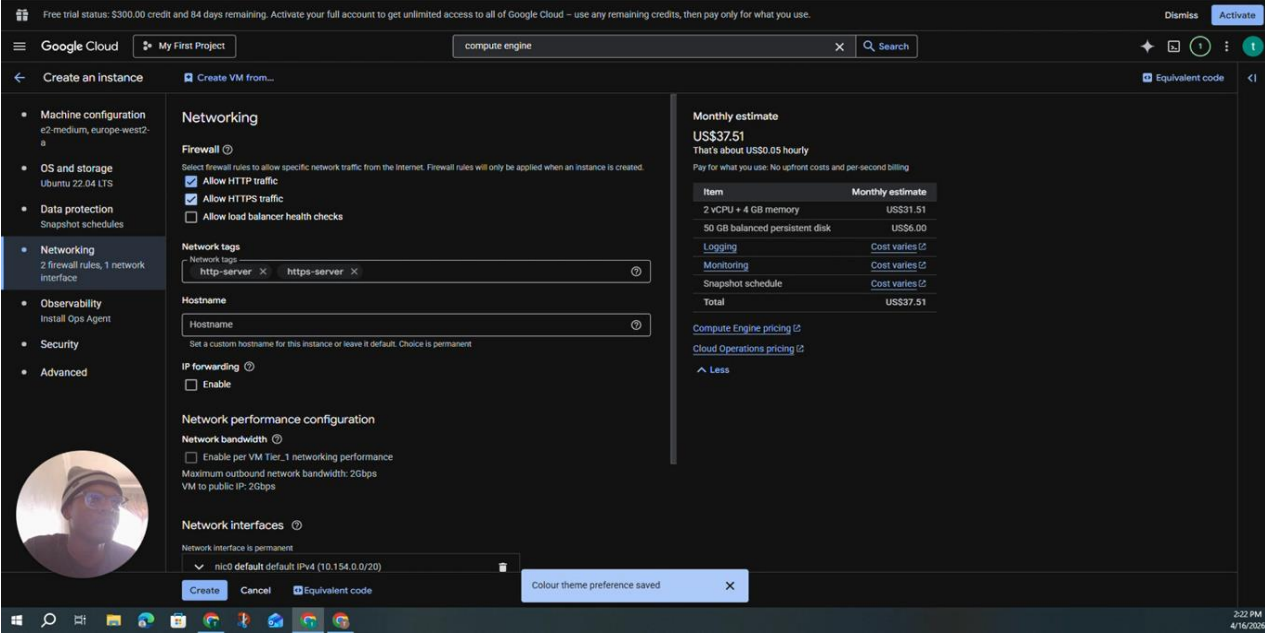


Figure 4

Figure 5: VM Instance Successfully Provisioned

VM Instances dashboard confirming that the agency-vm instance is running (green status indicator) in zone europe-west2-b with External IP: 35.197.228.25. A second stopped instance is also visible — this reflects the zone stockout error encountered during initial provisioning in europe-west2-a, which was resolved by switching to the available europe-west2-b zone. This demonstrates real-world cloud troubleshooting capability.

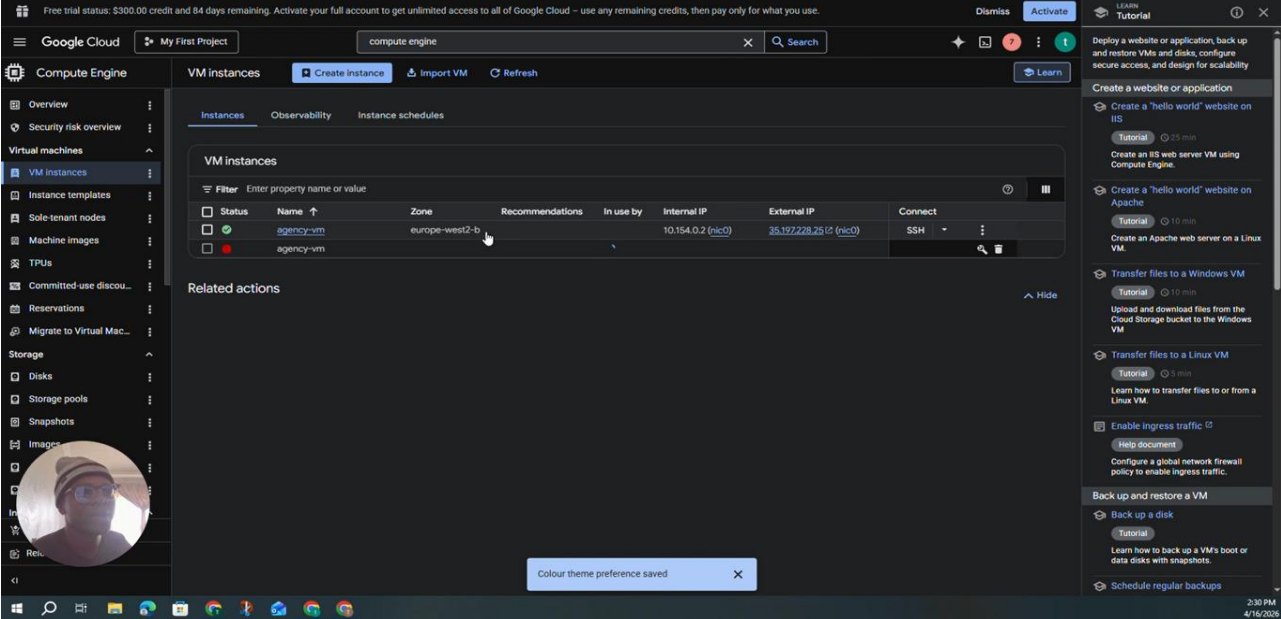


Figure 5

Note — Figures 6, 7, 9, 10, 11, 12, 13:

These captures represent active SSH-in-browser terminal sessions conducted via the GCP Cloud Console. The black display is a known screen recording artefact caused by GPU-accelerated rendering in browser-based terminals on certain display configurations. The corresponding commands executed during these sessions are fully documented in Figures 8, 14, and 15.

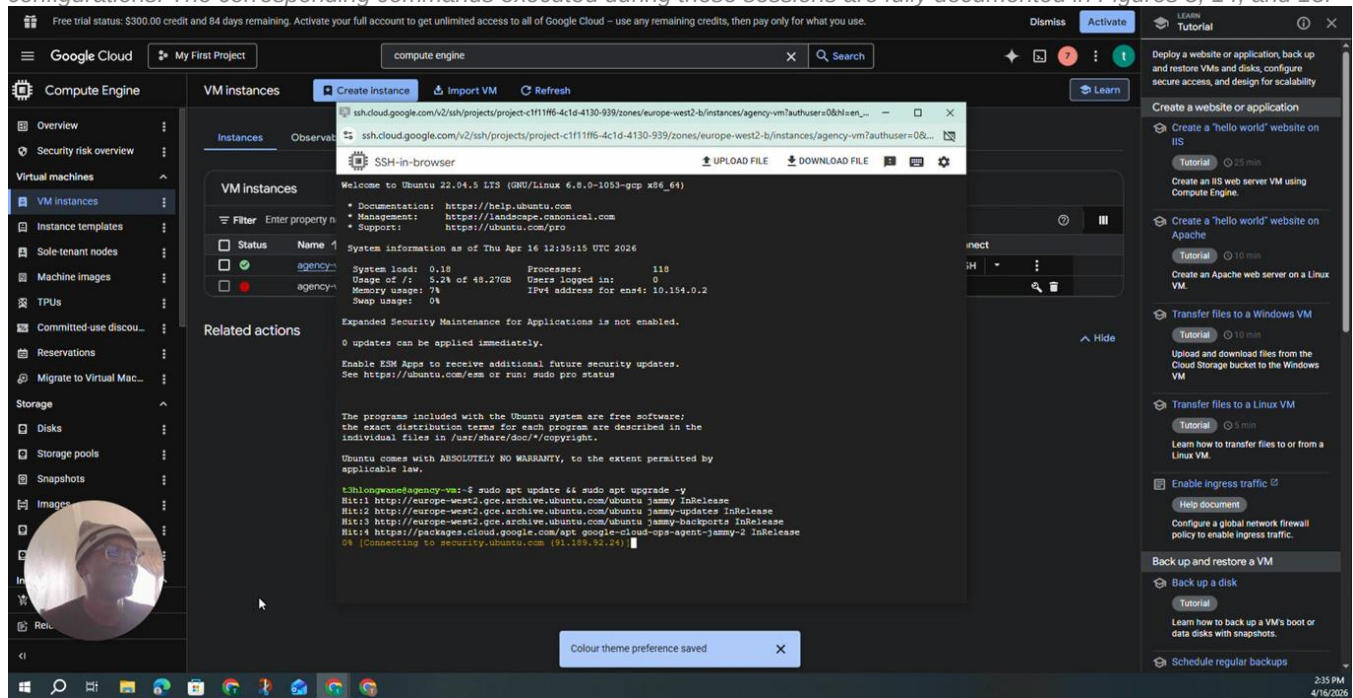


Figure 6: Connected to the VM and Updated the System.

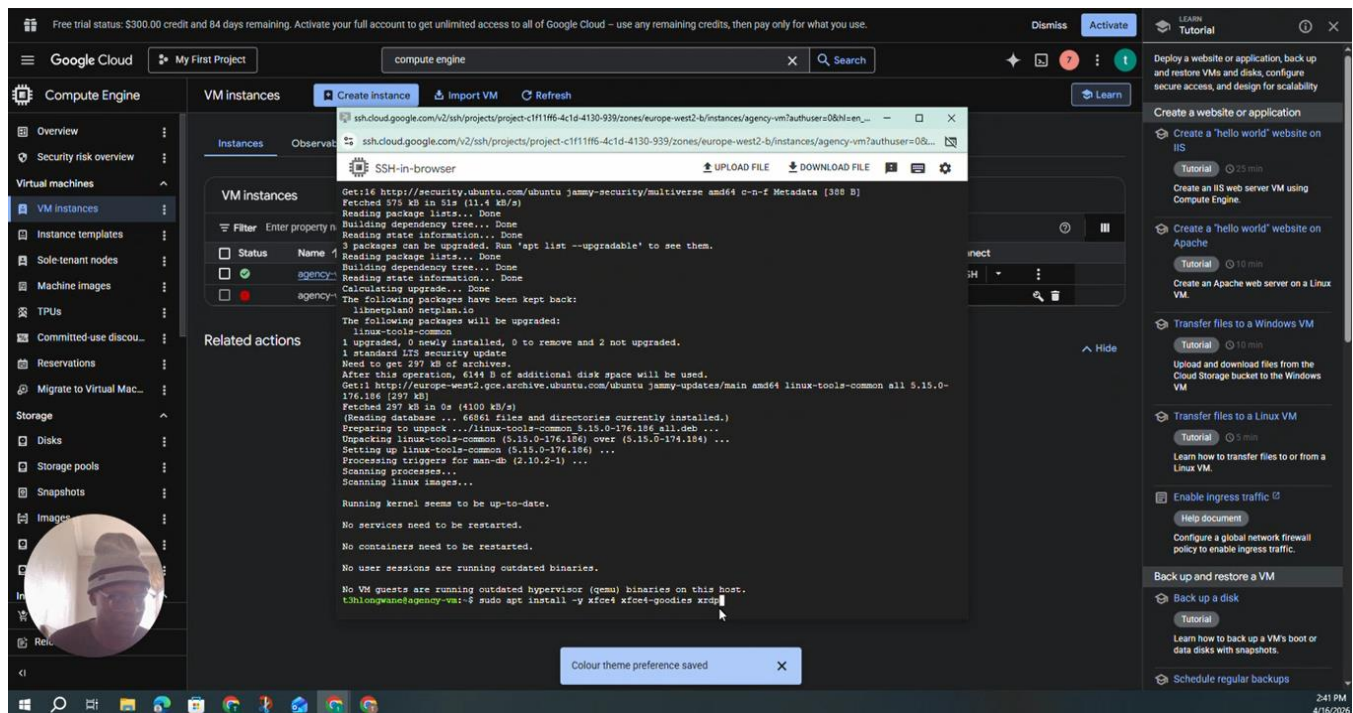


Figure 7: Installing the Desktop environment.

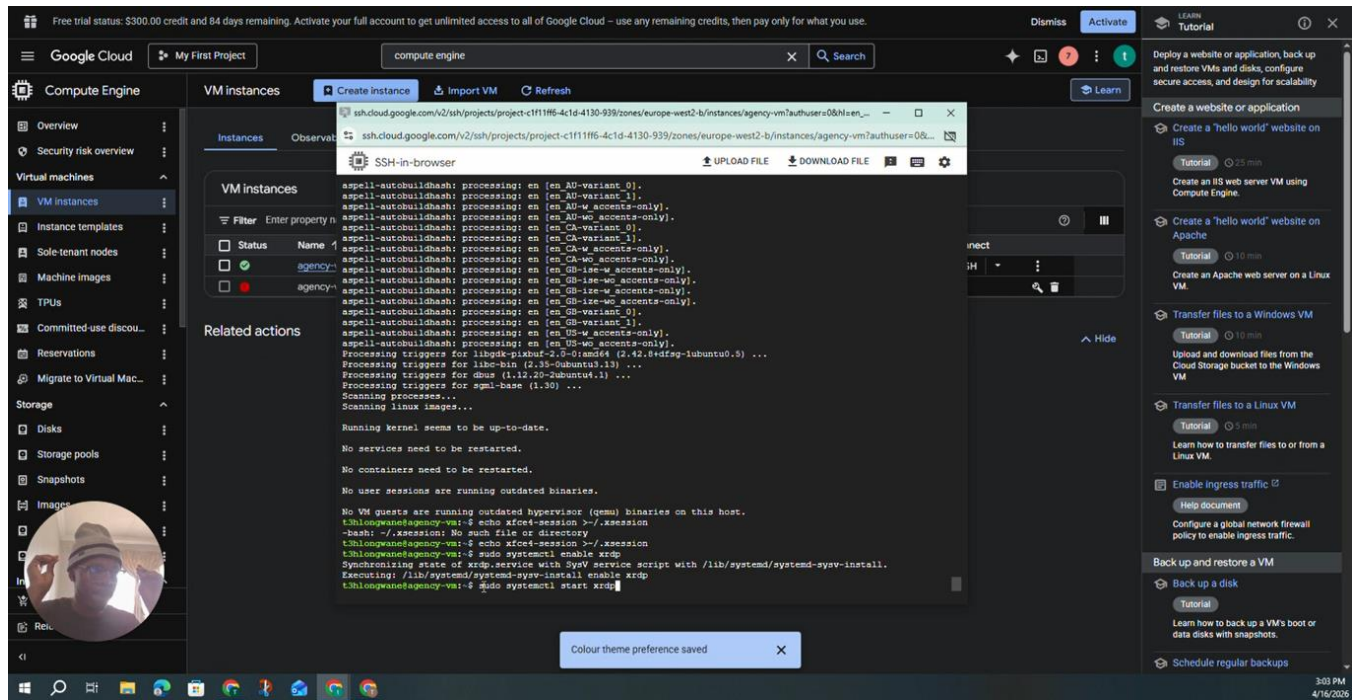


Figure 8: Configuring RDP.

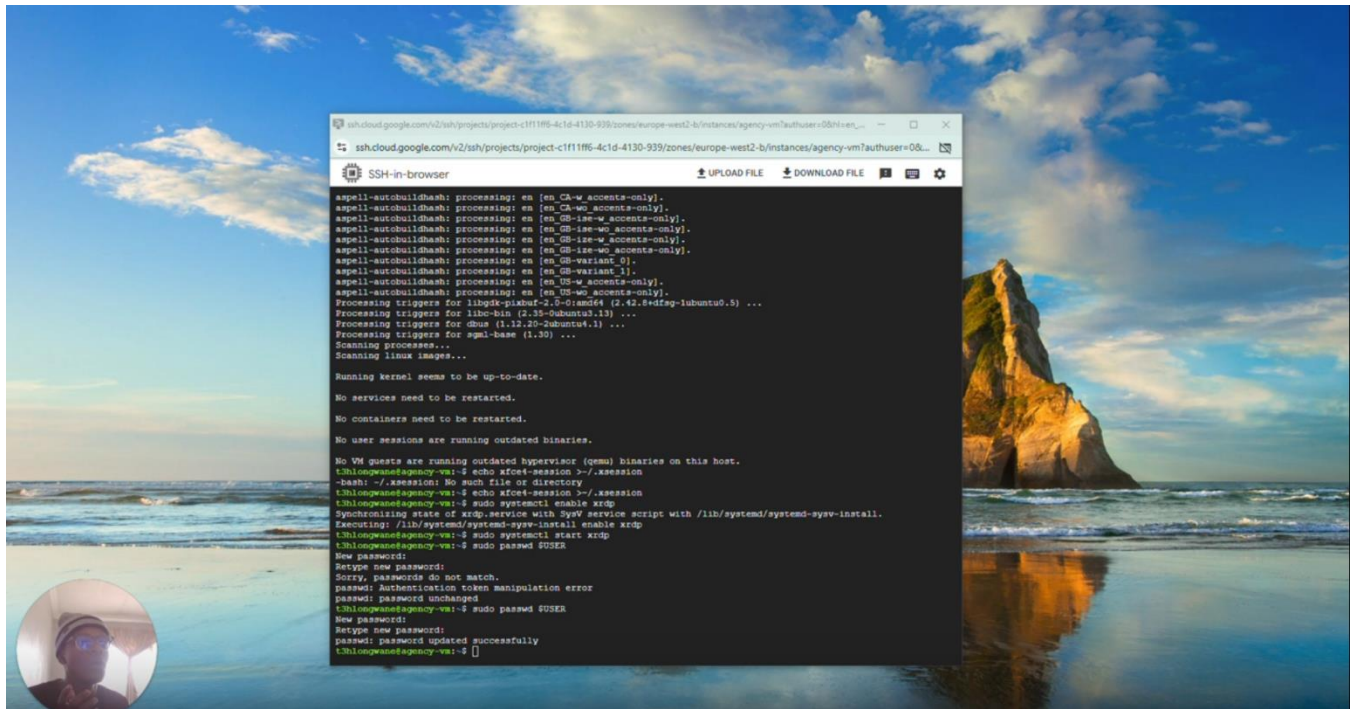


Figure 9: Setting User Password.

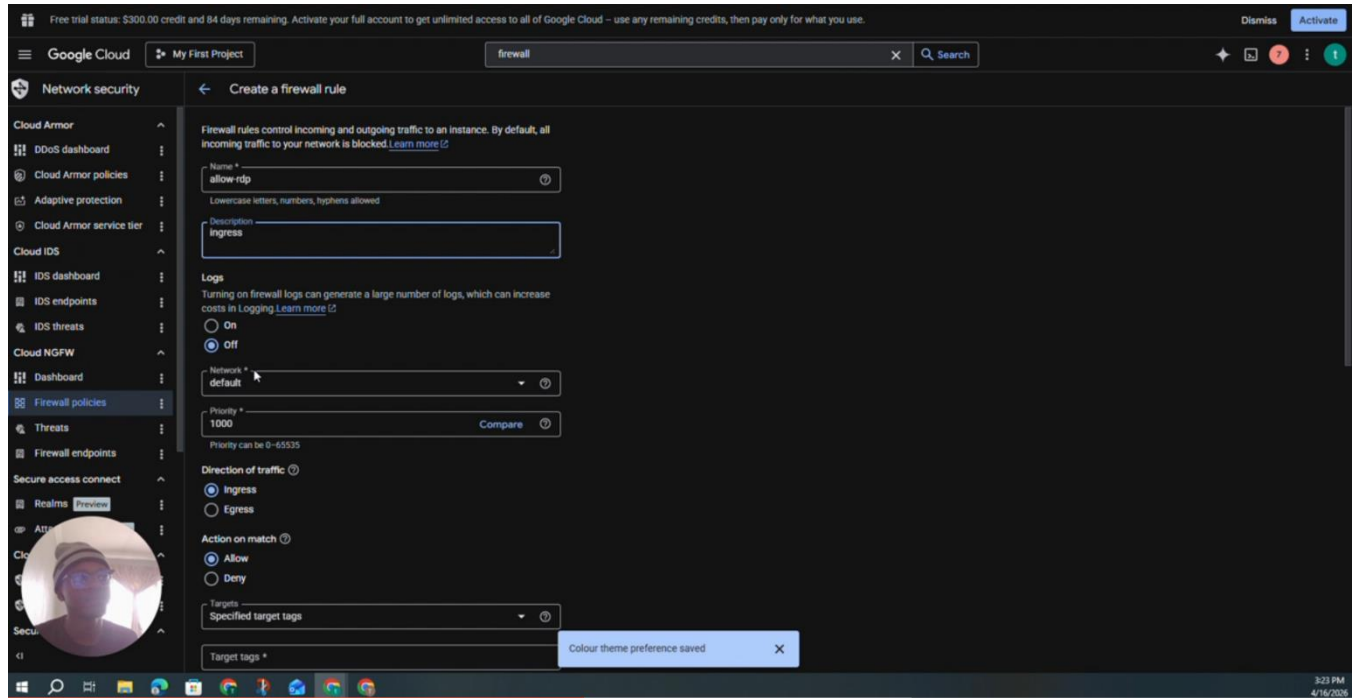


Figure 10: Configuring Firewall Policies for RDP.

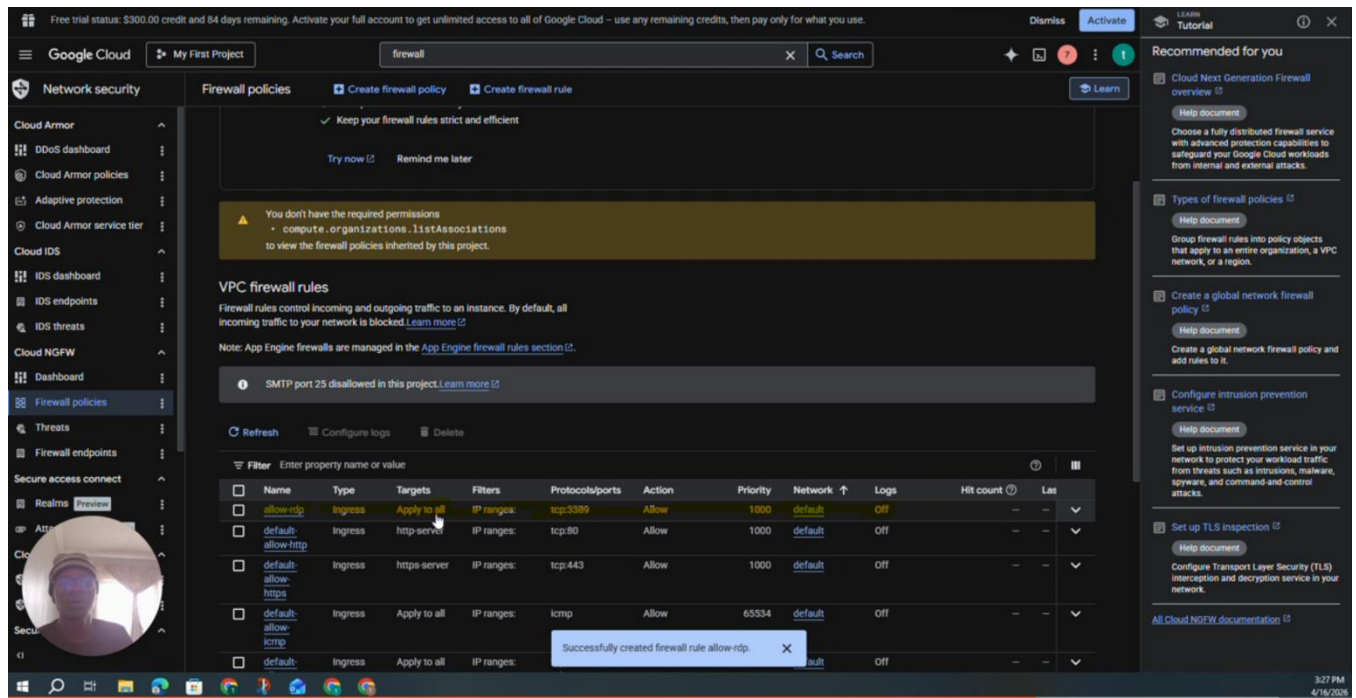


Figure 11: Firewall Settings Set.

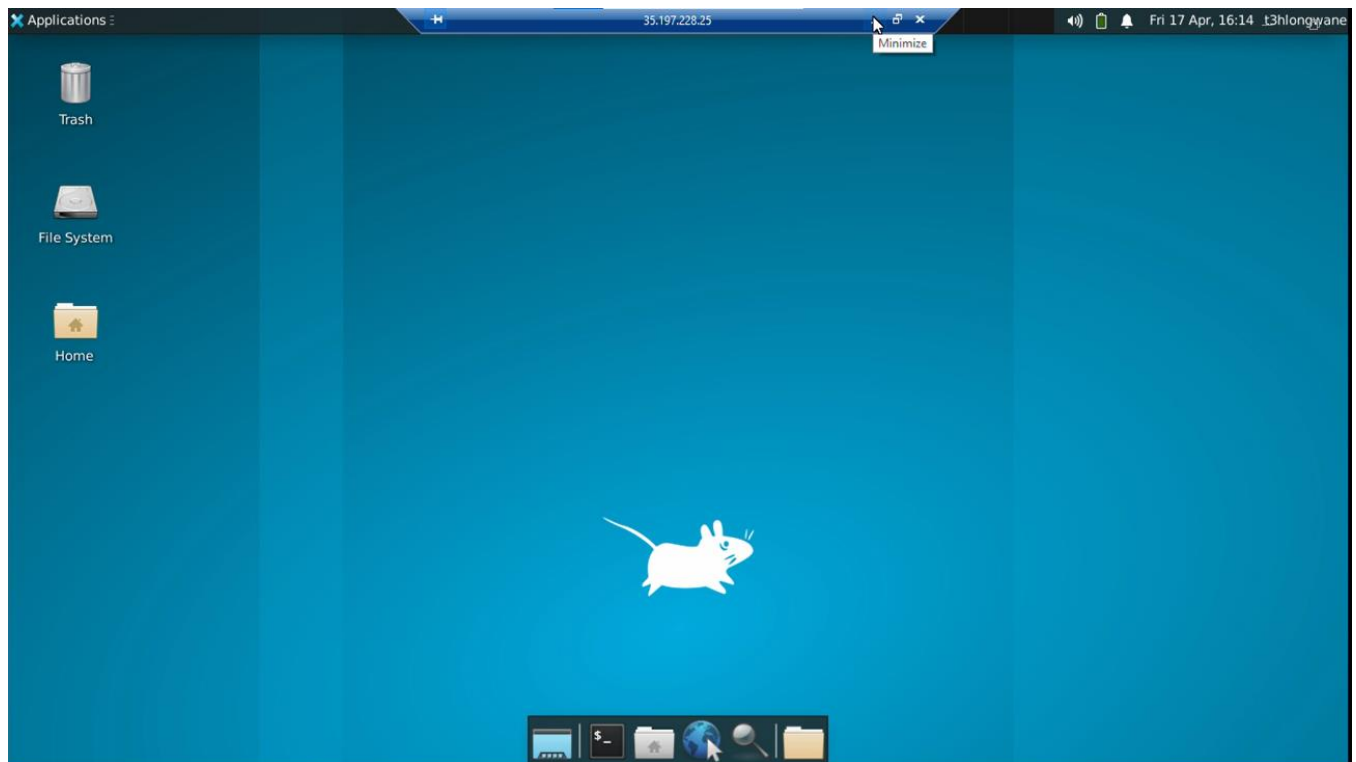


Figure 12: VM Running and live.

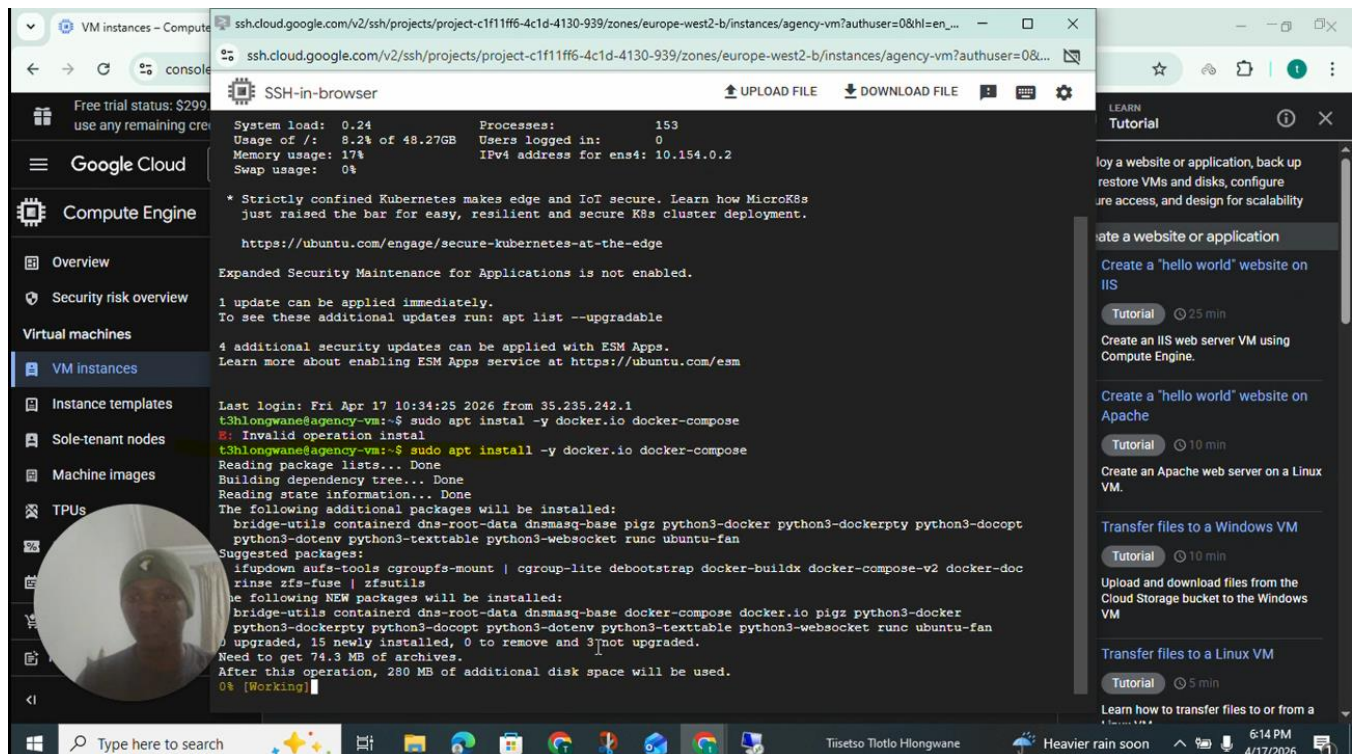


Figure 13: Installing Docker on VM.

Figure 8: Remote Desktop Environment Configuration

SSH terminal output showing the installation and configuration of the XFCE desktop environment and XRDP daemon. Commands executed include: `echo xfce4-session > ~/.xsession` (configuring the default desktop session for the user), `sudo systemctl enable xrdp` (registering XRDP as a persistent systemd service to survive reboots), and `sudo systemctl start xrdp` (starting the XRDP service immediately). This established the RDP endpoint on the VM.

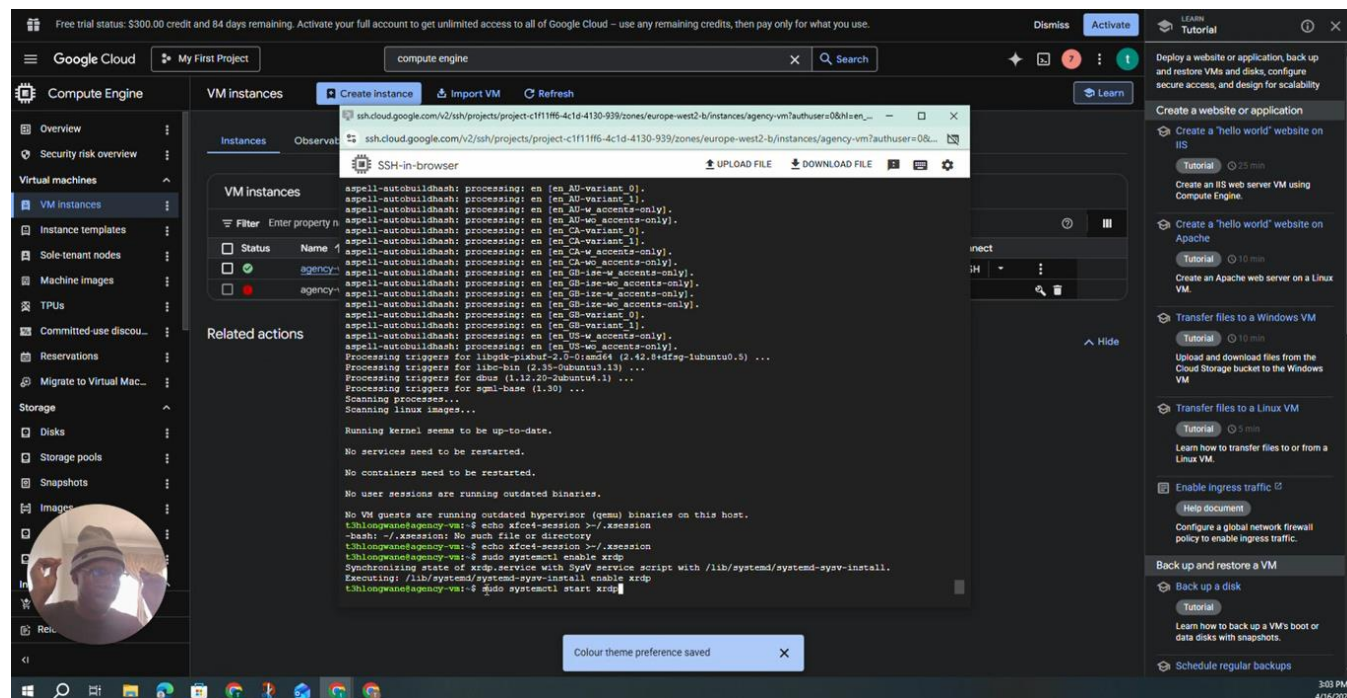


Figure 8

Figure 14: Docker Installation and N8N Container Deployment

N8N community edition successfully pulled and deployed as a Docker container. The docker run command configured the container with: detached execution mode (-d), automatic restart policy (--restart unless-stopped), persistent volume storage (n8n_data:/home/node/.n8n) to preserve workflow data across container restarts, and port mapping (5678:5678) to expose the N8N web interface. The subsequent docker ps command confirms the container is active with status 'Up 52 seconds', validating successful containerisation and runtime operation.

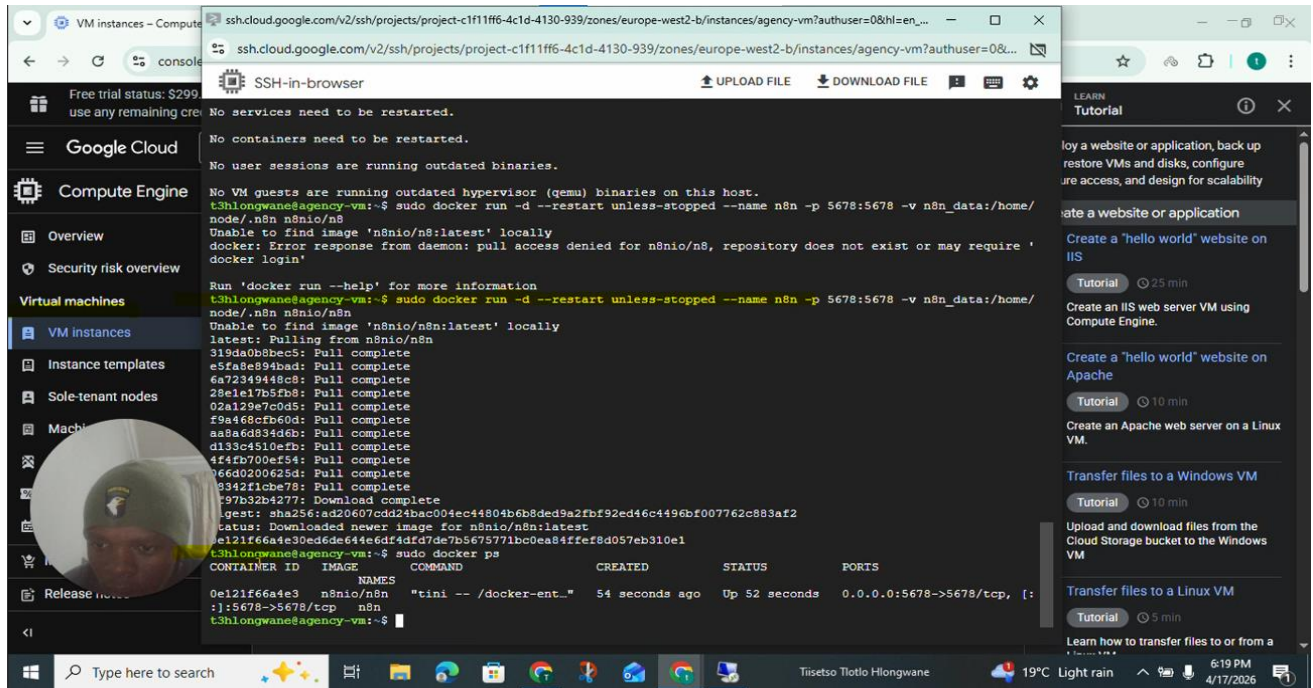


Figure 14

Figure 15: Firefox Browser Installation on the VM

Installing Firefox on the Ubuntu VM via the apt package manager to enable browser-based access to locally running services. The terminal output confirms successful download and installation of the Firefox snap package (149.0.2-1 from Mozilla). This was a prerequisite for accessing the N8N web interface from within the remote desktop session, as no GUI browser was pre-installed on the server OS.

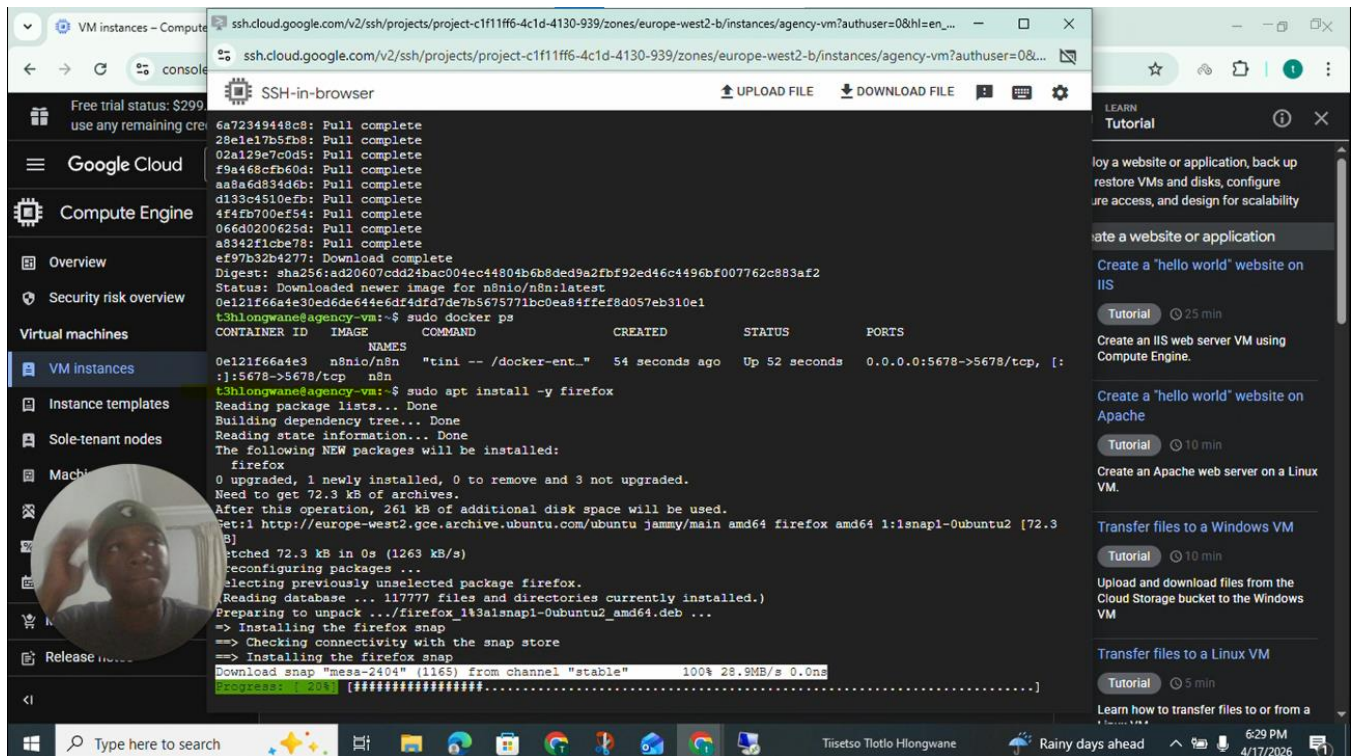


Figure 15

Figure 16: RDP Connection Initiated from Local Windows Workstation

Windows Remote Desktop Connection client establishing a connection to the cloud VM at IP address 35.197.228.25 over port 3389. This demonstrates successful network connectivity between the local Windows workstation and the cloud-hosted Ubuntu VM, validating the VPC firewall rule configured to allow TCP ingress on port 3389. The connection status 'Initiating remote connection' confirms that the XRDP daemon is accepting incoming RDP requests.

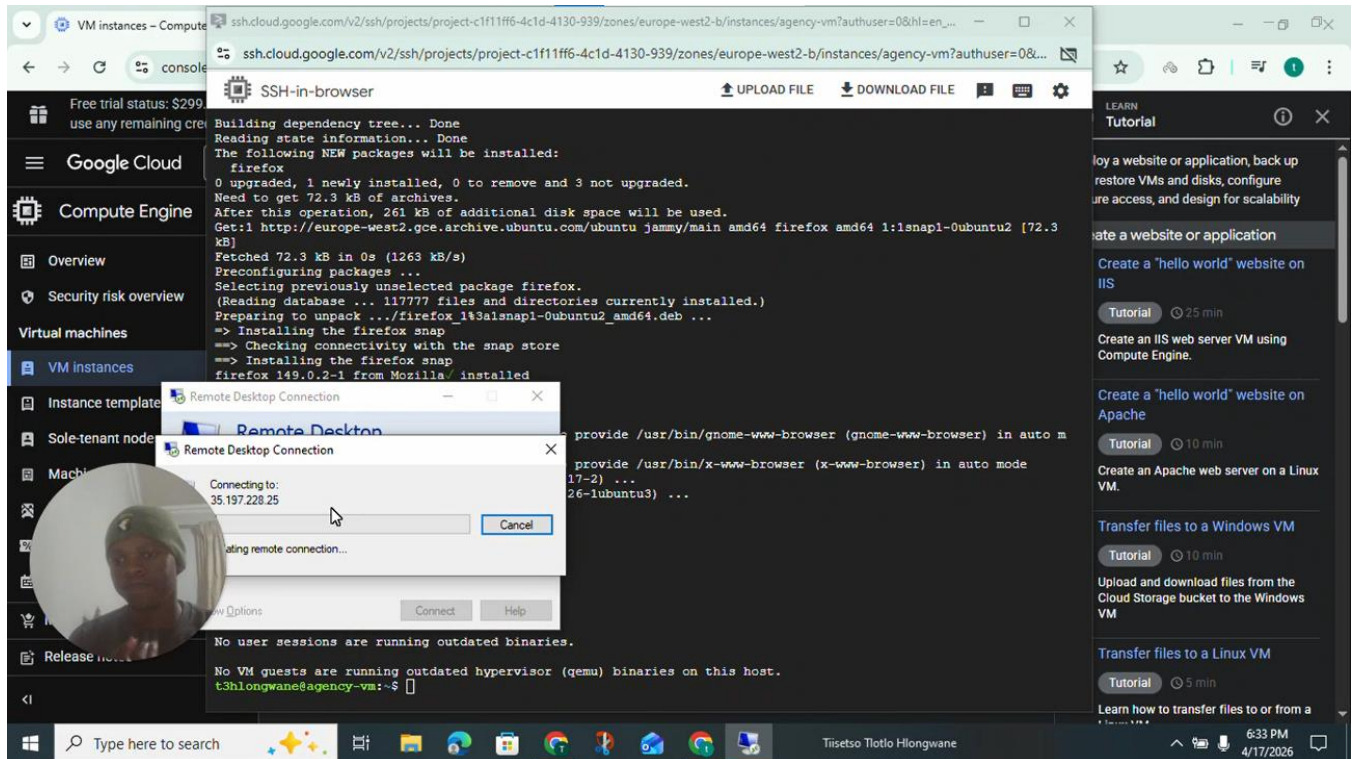


Figure 16

Figure 17: N8N Community Edition — Operational Validation

N8N workflow automation platform successfully running at `http://localhost:5678` within the active remote desktop session. The dashboard displays a fully authenticated instance with the user account created, community edition registered, and the platform ready for workflow development. The RDP session title bar confirms the remote IP (35.197.228.25) and the date/time of access. This figure validates the complete end-to-end deployment of the containerised application within the remote desktop environment.

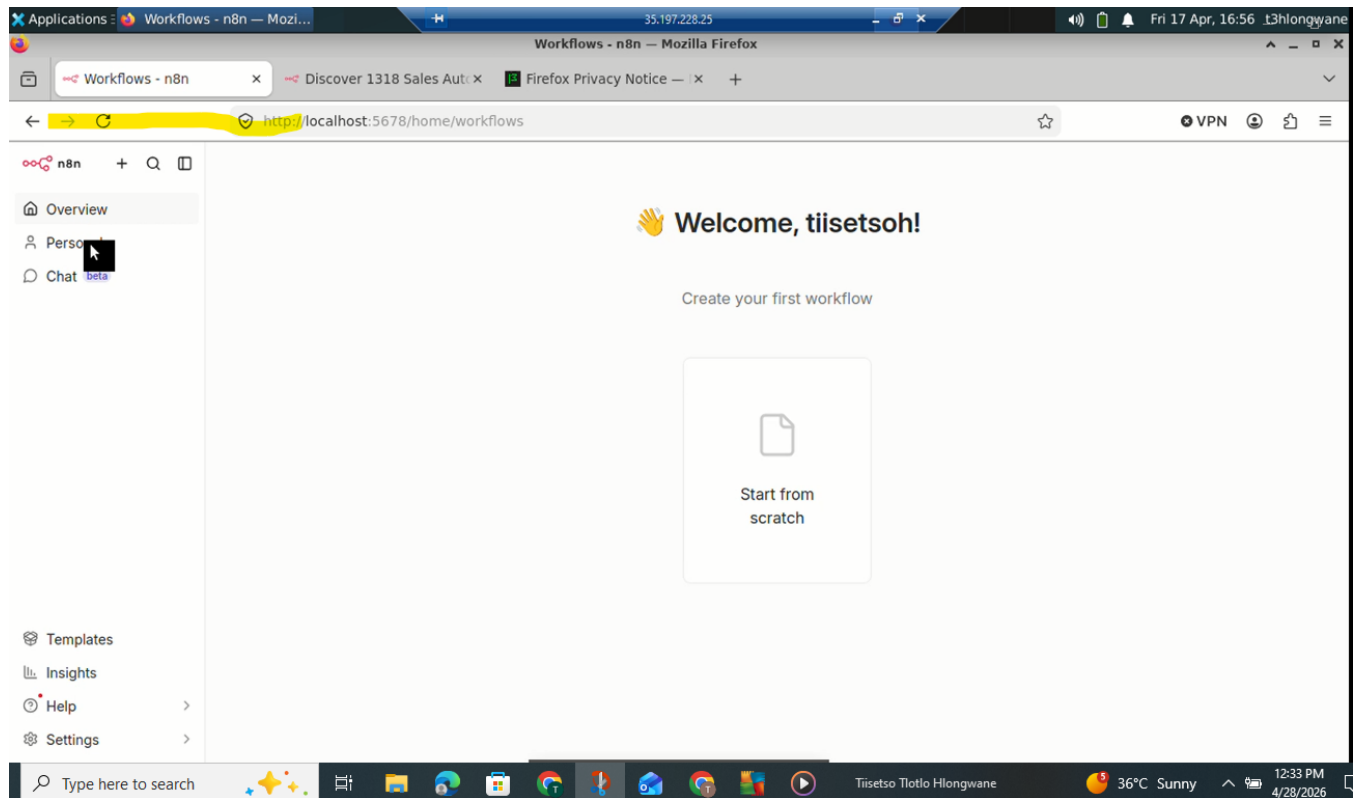
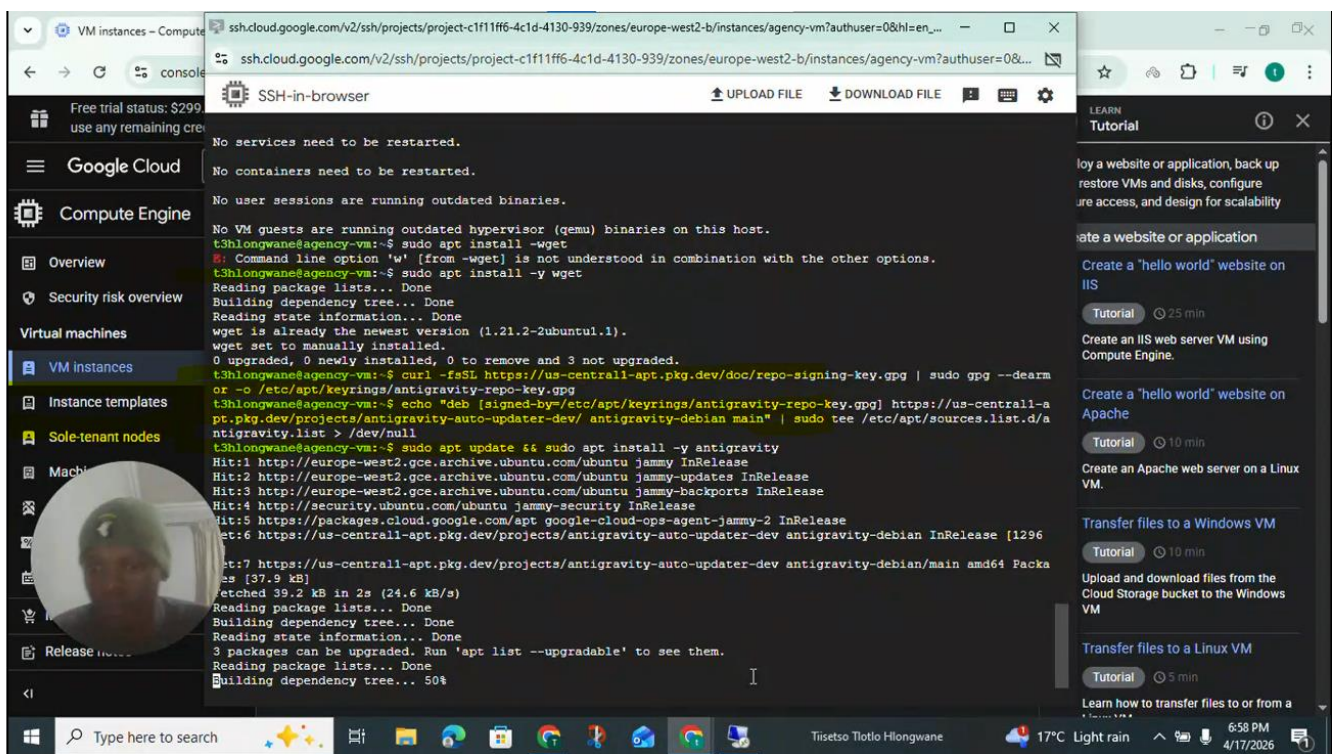


Figure 17

Figure 18: Antigravity IDE Installation via External APT Repository

Terminal output documenting the three-step installation process for the Antigravity IDE: (1) retrieval and registration of the GPG signing key from the Google Cloud APT repository endpoint (`us-central1-apt.pkg.dev`), (2) addition of the Antigravity Debian package source to the system APT sources list (`/etc/apt/sources.list.d/antigravity.list`), and (3) execution of `sudo apt update && sudo apt install -y antigravity`. This demonstrates third-party package management, GPG key handling, and secure repository configuration on a Linux server.



```
ssh.cloud.google.com/v2/ssh/projects/project-c1f11ff6-4c1d-4130-939/zones/europe-west2-b/instances/agency-vm?authuser=0&hl=en_...
SSH-in-browser
No services need to be restarted.
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
t3hlongwane@agency-vm:~$ sudo apt install -y wget
t3hlongwane@agency-vm:~$ sudo apt install -y wget
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
wget is already the newest version (1.21.2-2ubuntu1.1).
wget set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
t3hlongwane@agency-vm:~$ curl -fsSL https://us-central1-apt.pkg.dev/doc/repo-signing-key.gpg | sudo gpg --dearmor
or -o /etc/apt/keyrings/antigravity-repo-key.gpg
t3hlongwane@agency-vm:~$ echo "deb [signed-by=/etc/apt/keyrings/antigravity-repo-key.gpg] https://us-central1-apt.pkg.dev/projects/antigravity-auto-updater-dev/ antigravity-debian main" | sudo tee /etc/apt/sources.list.d/antigravity.list > /dev/null
t3hlongwane@agency-vm:~$ sudo apt update && sudo apt install -y antigravity
Hit:1 http://europe-west2.gce.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://europe-west2.gce.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://europe-west2.gce.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:5 https://packages.cloud.google.com/apt google-cloud-ops-agent-jammy-2 InRelease
Get:6 https://us-central1-apt.pkg.dev/projects/antigravity-auto-updater-dev antigravity-debian InRelease [1296 B]
Get:7 https://us-central1-apt.pkg.dev/projects/antigravity-auto-updater-dev antigravity-debian/main amd64 Packages [37.3 kB]
Fetched 39.2 kB in 2s (24.6 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... 50%
```

Figure 18

Figure 19: Antigravity IDE — Successful Authentication

The Google Antigravity authentication success page (antigravity.google/auth-success) confirms that the IDE was installed, launched, and authenticated within the cloud VM environment. The application is running natively on the XFCE remote desktop, completing the project objective of deploying both a containerised automation platform (N8N) and an AI-assisted development environment (Antigravity) on a single GCP virtual machine.

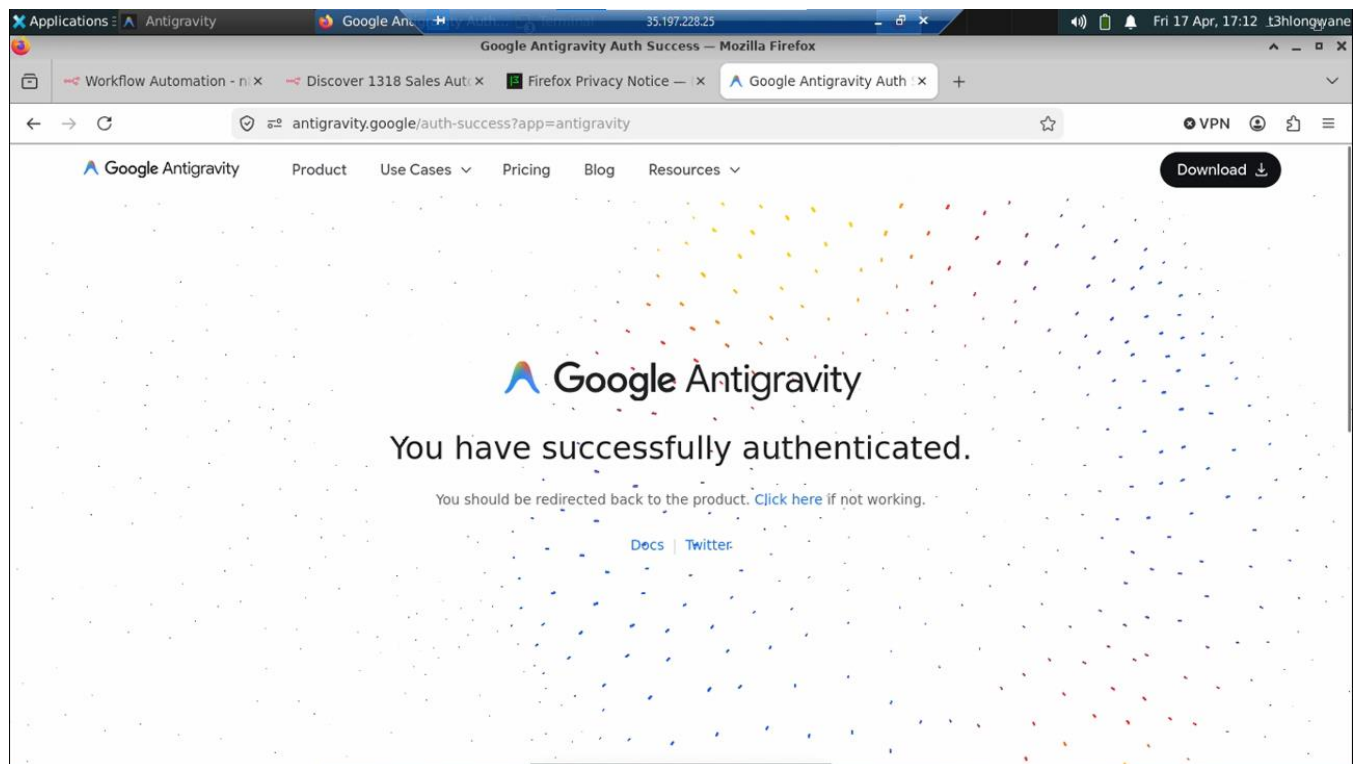


Figure 19

03

SKILLS & COMPETENCIES DEMONSTRATED

Cloud Engineering	Systems Administration	DevOps
· GCP VM provisioning · VPC networking · Firewall rule configuration · Region/zone selection · Cloud billing management	· Ubuntu 22.04 LTS admin · apt package management · systemd service control · User account management · Remote desktop (XRDP)	· Docker containerisation · Persistent volume storage · Port mapping & networking · Third-party apt repos · GPG key management

Competency Assessment: Associate Level

Beginner	✓ ASSOCIATE	Expert
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This portfolio of evidence documents the successful design and deployment of a fully functional cloud infrastructure environment on Google Cloud Platform, completed end-to-end as a practical, hands-on project.

The project required the application of competencies across three distinct IT disciplines simultaneously: cloud infrastructure engineering, Linux systems administration, and DevOps containerisation. Beginning from a blank GCP project, the candidate provisioned a production-grade compute instance, configured virtual private cloud networking and firewall rules, installed and configured a full graphical desktop environment accessible via the Remote Desktop Protocol, deployed a containerised automation application using Docker, and installed a third-party AI-assisted development environment from an external package repository.

Beyond the technical implementation, the project also demonstrated real-world problem-solving capability. When a zone stockout error was encountered during initial VM provisioning, the candidate identified the root cause and resolved it by migrating to an available availability zone with no loss of configuration. When SSH firewall connectivity was interrupted following a VM restart, the candidate diagnosed the missing firewall rule for port 22, applied the appropriate fix, and restored access — illustrating the kind of systematic troubleshooting expected of an associate-level infrastructure professional.

The project was executed on live cloud infrastructure using real credentials and billing resources, not a simulated environment — which strengthens the validity of this evidence as a demonstration of practical, deployable skill. The outcome is a cloud workstation that is actively usable for professional work: N8N community edition is operational and ready for workflow automation development, and Antigravity IDE is installed, authenticated, and available as a coding environment.

This project, combined with the candidate's existing certifications in Fortinet Cybersecurity (Fundamentals and Associate), Oracle Cloud Infrastructure, and hands-on experience with Microsoft Azure and Active Directory, demonstrates a well-rounded and multi-cloud capable infrastructure skill set at the associate level — with a clear trajectory toward senior cloud and systems engineering roles.

➤ **Video Links of the Project:**

➤ [Project Vid-3](#)

➤ [Project Vid-2](#)

➤ [Project Vid-1](#)